Do experiencers constitute a distinct role category within the event role repertoire?

Psycholinguistic evidence supports only two event role categories with distinct processing correlates (Rissman & Majid, 2019): *agent* and *patient* roles. However, there is no psycholinguistic evidence supporting the experiencer role. The proto-role approach (Dowty, 1991) aligns with this evidence as it proposes only two role categories: *proto-agent* and *proto-patient.* Hence, experiencers do not constitute a distinct role category but are grouped under the proto-agent role due to its entailment of sentience. Here, we investigate whether experiencer role exhibits specific processing correlates or not.

Previous studies showed an attentional preference to fixate on agents compared to patients (Gómez-Vidal et al., 2022), and a preference to transiently interpret the first unmarked noun in a sentence as an agent (Bornkessel & Schlesewsky, 2006; Isasi-Isasmendi et al., 2023; Sauppe et al., 2023). Based on this, we predicted that if experiencers fit into the proto-agent role, they would have parallel processing correlates as agents and different from patients. Otherwise, if the experiencer constitutes a specific category within the event role repertoire (Fillmore, 1971), we predicted that experiencers would show a specific processing pattern, different from both agents and patients.

We conducted an eye-tracking and an EEG experiments. In each experiment, 30 Spanish speakers read intransitive sentences with either agent (a), experiencer (b), or patient subjects (c). The same materials were used for both experiments: 96 experimental items and 96 fillers. An example of an experimental item is: (a) *La conductora*$_{AGENT}$ *el jueves se esforzó durante el examen*; (b) *La conductora*$_{EXPERIENCER}$ *el jueves se sorprendió durante el examen*; (c) *La conductora*$_{PATIENT}$ *el jueves se desorientó durante el examen* ("The driver on Thursday made an effort/was surprised/became disoriented during the exam"). We analyzed eye-tracking data with Bayesian Mixed-Effects Models and EEG data with Generalized Additive Mixed Models. In eye-tracking, agents and experiencers elicited longer regression path fixation times than patients in the post-verb region ($\hat{\beta}_{\text{Agent-Patient}}$ = -36.5, CrI = [-65.783, -6.7]; $\hat{\beta}_{\text{Experiencer-Patient}}$ = -70, CrI = [-102.082, -36.8]), higher regressions out counts in the same region ($\hat{\beta}_{\text{Agent-Patient}}$ = -0.0364, CrI = [-0.0697, -0.00231]; $\hat{\beta}_{\text{Experiencer-Patient}}$ = -0.0578, CrI = [-0.0940, -0.02320]), and higher regression in counts in the verb region ($\hat{\beta}_{\text{Agent-Patient}}$ = -0.0574, CrI = [-0.1068, -0.00749]; $\hat{\beta}_{\text{Experiencer-Patient}}$ = -0.0710, CrI = [-0.1231, -0.02037]). In ERPs, patients elicited a larger negativity over posterior electrodes compared to agents and experiencers, with no such negativity between agents and experiencers (Figure 1). This effect started in the early time-window and continued in the late time-window. Agents displayed a frontal negativity compared to experiencers in the early-time window (Figure 1). In the time-frequency analysis, patients elicited a power increase compared to agents and experiencers over central electrodes in theta band in the late time-window (Figure 2B, C), a power increase in alpha band over right-posterior electrodes in the early time-window (Figure 2E, F) and a power decrease in low-beta band over central and right-posterior electrodes in the early time-window (Figure 2H, I). Weaker differences were found when comparing agents and experiencers (Figure 2A, D, G), revealing distinct patterns than the one found when comparing these two to patients. Eye-tracking results revealed that agents and experiencers elicited an attentional preference compared to patients. ERP and TFA results showed that patients elicited a recategorization of the event role category due to the agent preference; and that this recategorization did not appear for experiencers. The subtle differences between agents and experiencers are best explained as proto-agent prototypicality effects rather than evidence for a distinct role category. This suggests that agents and experiencers are grouped under the category of proto-agent as both have proto-agent entailments (Dowty, 1991). Hence, the experience role does not seem to constitute a distinct role category in the syntax-semantics interface.

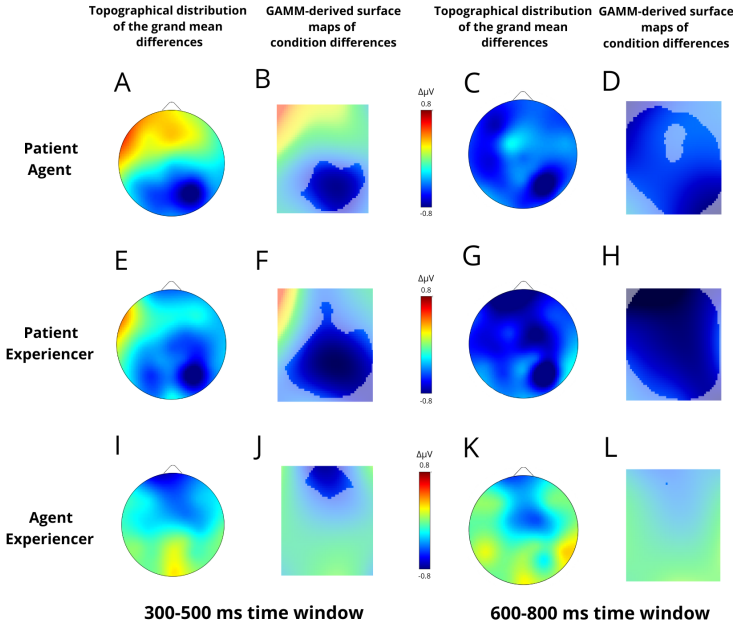

Figure 1. Pairwise comparisons of ERP differences between conditions at the verb region. In the GAMM plots, bright areas indicate differences whose 95% credible interval excludes zero.

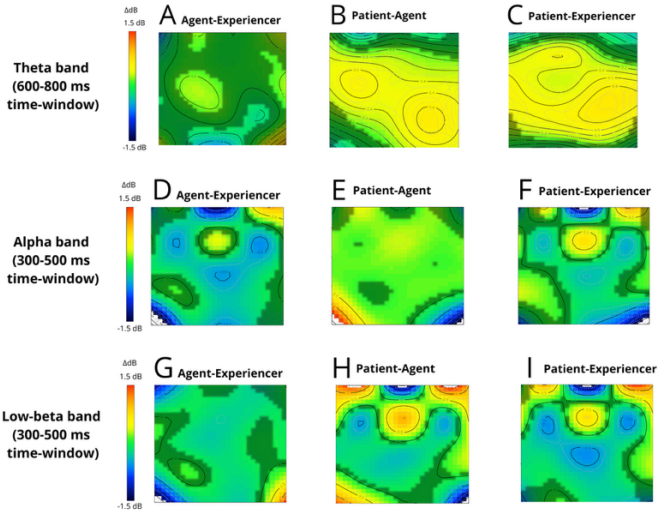

Figure 2. Pairwise comparison of power differences after verb region presentation. Non-shaded (bright) areas indicate significant differences.

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